

■ ListPlot3D

`ListPlot3D[array]` generates a three-dimensional plot of a surface representing an array of height values.

`ListPlot3D[array, shades]` generates a plot with each element of the surface shaded according to the specification in *shades*.

`ListPlot3D` returns a `SurfaceGraphics` object, which can be displayed using `Show`. ■ The following options can be given:

<code>AmbientLight</code>	<code>GrayLevel[0.]</code>	ambient illumination level
<code>AspectRatio</code>	<code>1</code>	ratio of height to width
<code>BoxRatios</code>	<code>{1, 1, 0.4}</code>	bounding 3D box ratios
<code>Boxed</code>	<code>True</code>	whether to draw the bounding box
<code>ClipFill</code>	<code>Automatic</code>	how to draw clipped parts of the surface
<code>DisplayFunction</code>	<code>\$DisplayFunction</code>	function for generating output
<code>Framed</code>	<code>False</code>	whether to draw a frame
<code>LightSources</code>	(see below)	positions and colors of light sources
<code>Lighting</code>	<code>False</code>	whether to use simulated illumination
<code>Mesh</code>	<code>True</code>	whether to draw a mesh on the surface
<code>Plot3Matrix</code>	<code>Automatic</code>	perspective transformation matrix
<code>PlotColor</code>	<code>True</code>	whether to plot in color
<code>PlotLabel</code>	<code>None</code>	a label for the plot
<code>PlotRange</code>	<code>Automatic</code>	range of values to include
<code>Shading</code>	<code>True</code>	whether to shade polygons
<code>ViewPoint</code>	<code>{1.3, -2.4, 2.}</code>	viewing position

■ *array* should be a rectangular array of real numbers, representing *z* values. There will be holes in the surface corresponding to any array elements that are not real numbers. ■ If *array* has dimensions $m \times n$, then *shades* must have dimensions $(m - 1) \times (n - 1)$. ■ The elements of *shades* must be either `GrayLevel[i]` or `RGBColor[r, g, b]`. ■ The default light sources used are the same as for `Graphics3D`. ■ See page 137. ■ See also: `Plot3D`.